HDOC DAY 67

GCD Using Recursive Euclidean Algorithm

Aim Find the greatest common divisor of two positive integers using a recursive function.

Algorithm

1. Read two positive integers a and b.

2. If b equals zero, return a (base case).

3. Otherwise call gcd(b, a % b) recursively.

4. Print the returned GCD.

C Program

```
#include <stdio.h>

int gcd(int a, int b)
{
    if (b == 0) return a;
    return gcd(b, a % b);
}

int main(void)
{
    int a, b;
    printf("Enter two positive integers: ");
    if (scanf("%d%d", &a, &b) != 2 || a <= 0 || b <= 0) {
        printf("Invalid input.\n"); return 1;
    }
    printf("GCD: %d\n", gcd(a, b));
    return 0;
}
```

Test Cases

Input	Expected result
12 18	6
17 13	1
48 18	6

Compilation and Execution (Mac Terminal)

```
clang -std=c99 -Wall -Wextra day67.c -o day67
./day67
```

Sample Output

```
Enter two positive integers: 48 18
GCD: 6
```

Important Note

Euclid identity: gcd(a,b) = gcd(b,a%b). The recursion ends when b = 0.

Result

The program was implemented using the required technique and produces the expected results for the given test cases.